

TERMS OF REFERENCE

Integration of IDSR system with weather and environmental data for climate-sensitive disease early warning systems

CONTRACT MODALITY	Individual Consultancy
DUTY STATION	Tanzania, Dar Es Salaam (remote with travels to Dodoma and Dar es Salaam)
DURATION OF CONTRACT	5 months

Purpose of Assignment

Background:

Climate change in Tanzania is increasingly affecting disease transmission patterns and contributing to outbreaks of climate-sensitive diseases such as cholera, malaria, dengue fever, Rift Valley fever and other vector-, water- and food-borne diseases. While the Ministry of Health (MoH) has scaled up the utilization of the electronic Integrated Disease Surveillance and Response (eIDSR) system and strengthened community and event-based surveillance for early detection, reporting, contact tracing and referrals of suspected cases, the current system largely relies on routine disease surveillance data and operates independently from weather and environmental information. This limits the ability of health authorities to anticipate outbreak risks and implement timely preventive actions.

Tanzania can strengthen epidemic preparedness by linking the MoH eIDSR/DHIS2 platform with Tanzania Meteorological Authority (TMA) climate, weather and environmental data to generate early warnings for climate-sensitive diseases. The MoH eIDSR platform reports immediately notifiable diseases, weekly priority disease data, event-based surveillance and outbreak management. TMA provides weather forecasts, seasonal climate outlooks, hydrometeorological services, climate services and warning products, and Tanzania has advanced national multi-hazard early warning efforts under the Early Warnings for All agenda.

Climate variability and extreme weather events including heavy rainfall, flooding, drought and heatwaves can increase the risk of outbreaks such as malaria, cholera, dengue, Rift Valley fever, diarrheal diseases and other epidemic-prone conditions. Tanzania's recent investments in early warning systems, meteorological infrastructure and climate services create an opportunity to connect climate intelligence with public health surveillance for earlier detection, forecast-based preparedness and timely response.

The MoH, in collaboration with UNICEF, TMA and relevant stakeholders, seeks to strengthen preparedness and response capacities for public health emergencies through the integration of meteorological and environmental datasets into the eIDSR/DHIS2 system. This initiative will develop climate-informed disease surveillance and early warning capabilities that can generate predictive alerts and inform public health decision-making at national, regional, council and facility levels

Purpose and Objective

Purpose

The purpose of this consultancy is to establish a climate-informed disease early warning system through

integration of epidemiological, meteorological and environmental data streams within, or interoperable with, the national eIDSR/DHIS2 platform. The consultancy will provide the technical foundation, interoperability framework, predictive analytics requirements, governance mechanisms, prototype configuration, and validation results required to operationalize climate-sensitive disease forecasting and early action within Tanzania's public health surveillance system

Objectives

To establish an integrated climate-informed disease early warning system by linking MoH eIDSR/DHIS2 surveillance data with TMA weather, climate and environmental information to improve outbreak prediction, preparedness and timely response to climate-sensitive diseases in Tanzania.

Specific Objectives

1. Integrate data systems between MoH eIDSR/DHIS2 and TMA climate information systems/platforms through agreed interoperability mechanisms.
2. Identify and prioritize climate-sensitive diseases for early warning modelling and pilot implementation.
3. Develop risk thresholds and predictive analytics requirements using epidemiological, meteorological, and environmental indicators.
4. Produce climate-health dashboards, bulletins and alert mechanisms for national, regional, council and facility-level action.
5. Strengthen preparedness and response planning based on forecast-based disease risk information.
6. Improve coordination among MoH, TMA, PO-RALG, NEMC, disaster management structures, academia, and technical/development partners.

Key Questions to be Answered by the Consultancy

1. Which meteorological and environmental variables show the strongest association with priority climate-sensitive diseases in Tanzania and should be prioritized for integration into eIDSR/DHIS2?
2. Which climate-sensitive diseases and geographical areas should be prioritized for pilot implementation?
3. What technical and operational architecture is required to integrate weather and environmental data into the existing eIDSR/DHIS2 system without compromising data quality, timeliness, privacy, confidentiality and usability?
4. How should early warning thresholds, alert triggers, validation steps and response protocols be defined and institutionalized within national and subnational surveillance and response structures?
5. What governance, data sharing and sustainability arrangements are required between MoH, TMA and other institutions to support long-term operationalization?

Methodology:

The consultancy will use a participatory, multisectoral and iterative methodology involving desk review of surveillance, climate and environmental information systems; key informant interviews; stakeholder consultations; joint technical design workshops; interoperability requirements analysis and use-case definition; configuration of data integration workflows; prototype development; user testing; validation exercises; refinement following testing; training; deployment support; and development of a phased national

scale-up roadmap.

Scope of Work

The consultancy will focus on the technical, analytical and operational integration of meteorological and environmental data systems with the national electronic Integrated Disease Surveillance and Response platform to enable climate-sensitive disease early warning functionality. The scope covers system assessment, data integration design, predictive analytics configuration, dashboard and alert workflow design, user testing, validation of early warning workflows, and scale-up planning within existing public health structures.

The consultant will work under the supervision of UNICEF Tanzania Health Specialists for Digital Health and Emergencies, in close collaboration with the Ministry of Health and the Tanzania Meteorological Authority.

Priority Data Streams for Integration

From MoH / eIDSR

- Weekly IDSR disease case reports
- Immediately notifiable disease alerts
- Event-based surveillance signals
- Outbreak Management System data
- Health facility and council-level disease trends
- Laboratory confirmation data where available
- Vaccination, WASH and vector-control intervention data where relevant

From TMA and Environmental Sources

- Rainfall amounts, rainfall forecasts and rainfall anomalies
- Temperature trends and heat stress indicators
- Humidity
- Flood forecasts and warnings
- Seasonal climate outlooks
- Drought indices
- Severe weather alerts
- Vegetation indices, water bodies, land surface temperature and other satellite-derived indicators where available

Suggested Disease-Climate Linkage Matrix

Disease / Condition	Relevant Climate or Environmental Triggers	Potential Early Warning Use
Cholera / acute watery diarrhea	Heavy rainfall, flooding, water contamination, poor sanitation	Pre-position ORS, cholera kits, water treatment supplies; strengthen WASH and risk communication.
Malaria	Rainfall, temperature, humidity, vegetation, stagnant water	Targeted vector control, commodities, community alerts, and malaria service readiness.
Dengue	Rainfall, temperature, urban water storage, vector breeding	Urban hotspot alerts, vector surveillance, risk communication, and environmental management.

Rift Valley fever	Heavy rainfall, flooding, livestock outbreaks, vegetation changes	Joint human-animal-environment surveillance and One Health preparedness.
Diarrheal diseases	Flooding, drought, unsafe water sources	WASH preparedness, community messaging and supplies pre-positioning.
Heat-related illness	Heatwaves, high temperature, humidity	Health facility alerts, preparedness guidance and protection of vulnerable populations.

Proposed System Architecture

A practical model should use interoperability rather than replacing existing systems:

1. MoH eIDSR/DHIS2 remains the core disease surveillance system.
2. TMA provides climate and weather datasets through agreed APIs, scheduled data exchange, secure file transfer, or data warehouse linkage.
3. A climate-health analytics layer combines epidemiological and climate/environmental data.
4. Dashboards generate risk maps, thresholds and alerts by region, council, ward or health facility catchment area where feasible.
5. Alerts feed into the MoH Public Health Emergency Operations Centre, Regional and Council Health Management Teams, TMA for early warning dissemination channels, and community health/risk communication platforms.

Key responsibilities:

1. Conduct Situational assessment and requirement gathering

The consultant will conduct a detailed assessment of the current IDSR architecture, including data flow, reporting frequency, interoperability capacity, and existing linkages with external systems. This will include mapping of system components at national and subnational levels, identifying gaps in real-time data exchange, and assessing readiness for integration with external climate and environmental data sources such as rainfall, temperature, humidity, vegetation indices, and flood risk datasets. A review of governance structures for data sharing between the Ministry of Health, meteorological authorities, and other relevant institutions will also be undertaken.

Specifically, the consultant will:

- Review existing eIDSR/DHIS2 architecture, data flows, reporting tools, reporting frequency and climate-sensitive disease surveillance processes.
- Review available meteorological and environmental datasets, including rainfall, temperature, humidity, flooding, drought, vegetation and other relevant indicators.
- Map relevant TMA systems, datasets, forecast products, data access arrangements and potential integration points, including the National Meteorological Database and early warning systems where applicable.
 - o Conduct stakeholder mapping and consultations with MoH, TMA, PO-RALG, NEMC, disaster management structures, academia and partners.
- Document business requirements, user stories/use cases, interoperability requirements, governance arrangements, data-sharing requirements, privacy and confidentiality considerations.

2. System design and Integration Framework

- Identify priority climate-sensitive diseases and associated environmental triggers for surveillance and

modelling.

- Develop a Climate-Informed Disease Early Warning Framework and disease-climate linkage matrix
- Design the interoperability architecture between eIDSR/DHIS2 and TMA climate information systems/platforms.
- Define climate-health risk indicators, alert thresholds and predictive analytics requirements.
- Develop specifications for climate-health dashboards, risk maps, bulletins, and automated alerting workflows.
- Define data standards, metadata requirements, data quality checks, governance mechanisms, and operational workflows.

3. Stakeholder engagement and technical facilitation.

- Facilitate at least two physical working sessions and multiple virtual consultations.
- Build consensus on data standards, governance mechanisms, and operational workflows.
- Establish or strengthen a multisectoral technical working group.
- Prepare agendas, technical briefs, presentations, and meeting reports for each consultation.

4. Prototype development and user testing support

- Configure and test data integration workflows using training/test environments before production deployment.
- Configure climate-health dashboards, risk maps, alert workflows and bulletin templates.
- Validate alert thresholds using historical disease and climate data where available.
- Lead user acceptance testing and validation of exercises involving MoH, TMA, and subnational users.
- Document test results, lessons learned, and recommended improvements.

5. System deployment, Scale up and sustainability planning.

- Deploy the agreed solution to a production environment ready for operational use, subject to MoH/TMA approvals and ICT governance requirements.
- Develop a phased national implementation roadmap, including pilot regions, priority diseases, institutional arrangements and scale-up pathway.
- Recommend capacity-building needs, maintenance arrangements, data governance mechanisms and sustainability measures.
- Propose monitoring indicators for climate-informed surveillance performance and long-term operationalization

Expected Outputs

- MoH–TMA climate-health data-sharing framework.
- Integrated climate-health surveillance, architecture and technical specifications.
- Climate-sensitive disease risk maps and dashboard prototype.
- Disease-specific early warning thresholds and alert workflow definitions.
- Routine climate-health bulletin template.
- Validated disease–climate linkage matrix.
- User acceptance testing report and improvement plan.
- National scale-up roadmap and sustainability plan.
- Monitoring framework for climate-informed surveillance performance.

Suggested Coordination Structure

Lead Institutions

- Ministry of Health: disease surveillance, eIDSR/DHIS2 governance, public health response and alert

validation.

- Tanzania Meteorological Authority: weather, climate, hydrometeorological and early warning data and technical interpretation of meteorological information.
- UNICEF Tanzania: Technical assistance, coordination support, quality assurance and contract management.

Key Collaborators

- PO-RALG
- Disaster Management Department / Prime Minister's Office
- NEMC and relevant environmental agencies
- Ministry of Livestock and Fisheries for zoonotic diseases and One Health linkage
- Academic institutions, including DHIS2 technical partners
- WHO, WMO, FAO, CDC, World Bank, UNDP and other technical/development partners as relevant

Work Assignments Overview	Deliverables/Outputs	Delivery deadline to complete the deliverable
Provide detailed methodology, stakeholder engagement plan and approach, and timelines	Inception report and Workplan	15 August 2026
Conduct comprehensive situational assessment on existing IDSR systems, climate-sensitive disease surveillance processes, available meteorological and environmental datasets, stakeholder mapping, interoperability requirements, governance arrangements, and data sharing.	Structured report on Situational assessment, that should include findings, gaps, opportunities, and prioritized requirements for climate-informed disease surveillance integration.	30 August 2026
Develop and submit a climate-informed disease surveillance framework, including identification of priority climate-sensitive diseases and environmental triggers	Climate-informed disease early warning (surveillance) framework	10 September 2026
Facilitate stakeholders' consultation and working sessions to produce and validate system design and integration recommendations covering data standards, governance mechanism, operational flows,	System designed package with technical manuals, in training instances including dashboard	15 October 2026

alerts thresholds, risk indicators, predictive analytics requirement and proposed data integration architecture.		
Facilitate User Acceptance Testing (UAT) and validation exercises, and document lesson learned and recommended system improvement.	User acceptance testing report with lesson learned, and recommended system improvement	30 October 2026
Develop a phased national implementation roadmap, including institutional arrangements, capacity building requirements, sustainability measure, monitoring and evaluation indicators and recommendation for long term operationalization of climate-informed disease surveillance.	Enhancement of integration solution after UAT, Integration roadmap with proposed indicators for monitoring climate-informed surveillance performance and a dashboard showing these indicators with GIS maps	10 November 2026
Submit a final consultancy report summarizing achievements, key decisions, and recommendations.	Final comprehensive consultancy report	30 November 2026

Payment Schedule:

Payment will be made on submission and acceptance of deliverables. UNICEF reserves the right to withhold payment in case the deliverables submitted are not up to the required standard or in case of delays in submitting the deliverables on the part of the consultant.

Important Notes

- Individuals engaged under a consultancy or individual contract will not be considered “staff members” under the Staff Regulations and Rules of the United Nations and UNICEF’s policies and procedures and will not be entitled to benefits provided therein (such as leave entitlements and medical insurance coverage). Their conditions of service will be governed by their contract and the General Conditions of Contracts for the Services of Consultants and Individual Contractors. Consultants and individual contractors are responsible for determining their tax liabilities and for the payment of any taxes and/or duties, in accordance with local or other applicable laws.
- The selected candidate is solely responsible for ensuring that the health insurance (and visa if applicable) required to perform the duties of the contract are valid for the entire period of the contract. Selected candidates are subject to confirmation of fully-vaccinated status against SARS-CoV-2 (Covid-19) with a World Health Organization (WHO)-endorsed vaccine, which must be met prior to taking up the assignment. It does not apply to consultants who will work remotely and are not expected to work on or visit UNICEF premises, programme delivery locations or directly interact with communities UNICEF works with, nor to travel to perform functions for UNICEF for the duration of their consultancy contracts.
- UNICEF offers reasonable accommodation for consultants with disabilities. This may include, for example, accessible software, travel assistance for missions or personal attendants. We encourage you to disclose your disability during your application in case you need reasonable accommodation during the selection process and afterwards in your assignment.

**QUALIFICATIONS / SPECIALIZED KNOWLEDGE / EXPERIENCE/ COMPETENCIES
(CORE/TECHNICAL/FUNCTIONAL) / LANGUAGE SKILLS REQUIRED FOR THE ASSIGNMENT****Education:**

- Masters Health Informatics, Information System, Public Health, Epidemiology, Environmental Health, Climate Science, computer science or related fields.

Work Experience:

- Minimum 7 years’ experience of relevant professional experience in public health surveillance, health information systems, epidemiology, climate services, or digital health.
- Demonstrated experience with IDSR, DHIS2, disease surveillance platforms or health information system integration.
- Experience in climate-health data integration, early warning systems, or environmental health surveillance
- Proven experience facilitating multisectoral technical consultations and workshops
- Experience working with government institutions and development partners in Tanzania or similar contexts is an asset
- Demonstrated experience in interoperability architecture, APIs, data exchange mechanisms, data warehousing or integration of external datasets into digital health platforms.

- Experience in climate-health data integration, early warning systems, environmental health surveillance, disease modelling or predictive analytics

Language Requirements:

- Fluency in English with proven strong writing and presentation skills.

Knowledge/Expertise/Skills:

- Health information systems architecture and interoperability
- Data governance and data-sharing frameworks
- Climate and environmental data application for public health
- GIS, predictive analytics, and disease modelling knowledge is desirable
- Excellent analytical, report writing and facilitation skills
- DHIS2 configuration, integration and dashboarding knowledge.